Fitting the luminosity-function shape

Conditional likelihoods with and without measurement error in flux and distance

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survey_debias project note

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Abstract

This note records how we fit the *shape* of a TNO luminosity function (LF) from a debiased pencil-beam survey, and why the likelihood looks the way it does. The fit described here is deliberately *shape only*: it does not use the Horvitz–Thompson sum $\sum 1/P$ as a Poisson mean. Absolute normalisation is handled separately. I first write down the no-measurement-error case (catalogue H treated as truth), then the version that folds in photometric scatter from the survey's `.eff` files and, when available, per-object distance uncertainty. The aim is that someone who is comfortable with LF work, but not with quadrature jargon, can see what is being averaged over and what is not. Figures are generated by `docs/lf_shape_fit/make_figures.py`, which calls the production likelihood in `survey_debias.hfit`. A short results section applies it to the twelve dynamically cold objects of JWST Sample A.

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1 What problem we are solving

A survey hands us a list of detections with apparent magnitudes and (approximate) distances. From those we form absolute magnitudes H, and we want the differential luminosity function

$$\lambda(H) \equiv \frac{\mathrm{d}N}{\mathrm{d}H} \tag{1}$$

of the underlying population. Two things get in the way:

1. **Selection.** Faint objects are less likely to be found. The survey efficiency, tracking fraction, and characterisation cut are known functions of apparent magnitude (and sometimes rate of motion). Call the product of those the selection function $S(m)$. When a detection has to be found at several epochs, S is the product of the per-epoch selections.

2. **Measurement error.** The magnitude we wrote in the catalogue is not the true magnitude, and the distance we used to turn m into H is not exact. Both move the inferred H.

This note is about fitting the *shape* of $\lambda(H)$—the faint-end slope α and the bright-end taper (β, H_B)—while conditioning on the fact that each object was detected. We are not, in this document, fitting the total number of objects in the belt. That normalisation is a separate Horvitz–Thompson (or later, extended Poisson) step.

The functional form we use is the OSSOS exponentially tapered form (Kavelaars et al. 2021; Petit et al. 2023):

$$\ln N(< H) = (\ln 10)\,\alpha\,(H - H_o) - 10^{-\beta(H-H_B)}, \tag{2}$$

with $\lambda(H) = \mathrm{d}N/\mathrm{d}H$. H_o is a reference magnitude that cancels in a shape-only fit; we keep it only so that the cumulative form matches the literature.

Which end the parameters describe. Recall that H gets *smaller* as objects get *larger*: the bright end of the LF is the large-object end. The local logarithmic slope of equation (2) is

$$\frac{\mathrm{d}\log_{10} N(< H)}{\mathrm{d}H} = \alpha + \beta\, 10^{-\beta(H-H_B)}. \tag{3}$$

For $H \gg H_B$ (faint, small objects) the second term vanishes and the slope is α: α **is the faint-end slope**. For $H < H_B$ (bright, large objects) the second term grows without bound and $N(< H)$ is driven to zero: **the taper removes the bright, large-object end**. H_B is where that cut-off sets in and β how sharply it does so. Two consequences matter below:

- The local slope is never below α and only increases brightward, so the tapered form can only describe a distribution that gets *flatter* toward faint H.

- α is only measured if detections extend well faintward of H_B, and the taper only if they extend brightward of it. If H_B lies faintward of every detection, α describes magnitudes with no data.

2 Why a conditional likelihood

A natural first instinct in LF work is a weighted histogram or a weighted likelihood $\sum_k w_k \log f(H_k \mid \theta)$ with $w_k = 1/P_k$. That is fine for some purposes, but it mixes shape and normalisation, and it treats $\sum w_k$ as if it were an independent Poisson count. For the shape of $\lambda(H)$ I prefer a likelihood that asks a narrower question:

> *Given that object k was detected, and given its geometry (distance, field, rate), what is the probability density of the magnitude we observed?*

That is a *conditional* density. Write D_k for the event "detected in this survey under geometry k". The contribution of detection k is

$$f_k = p(\text{observed magnitude of } k \mid D_k, \text{geometry}_k, \theta). \tag{4}$$

The joint shape likelihood is $\prod_k f_k$. Absolute rate information has been conditioned out: two surveys with the same efficiency curve but different areas give the same shape likelihood for the same detections.

This is the same philosophy as a detection-conditioned LF fit in a magnitude-limited sample: the selection function appears in both the numerator and the denominator, and the overall survey yield cancels.

3 No measurement error

3.1 Setup

Ignore photometric and distance error for a moment. Then each detection has a known true apparent magnitude m_k and a known true absolute magnitude

$$H_k = m_k - \mu_k, \tag{5}$$

where μ_k collects the distance modulus, colour term, and phase correction. (For JWST Sample A, m is NIRCam F150W2, H is in r with $r - \text{F150W2} = 1.0$, and $r = \Delta = d_{\text{bary}}$ with the Bowell $G = -0.12$ law, as in `grid_bias.apparent_to_Hr`.)

Detection probability at magnitude m is the selection function

$$S(m) = \prod_e \eta_e(m)\, \text{track}_e(m)\, \text{char}_e(m), \tag{6}$$

read from the block's `.eff` file at each epoch e: η is the detection probability, track is the tracking fraction, and char is the characterisation cut (`mag_lim`, or $\eta \geq 0.4$ when no limit is set). The product over epochs is what OSSSSim applies when `survey.conf` requires a detection at every epoch. For Sample A, which requires all three, S is the cube of the single-epoch efficiency (Figure 1).

3.2 Likelihood

Under a population $\lambda(H)$, the joint density of "draw an object of absolute magnitude H and detect it at this geometry" is proportional to $\lambda(H)\, S(H + \mu_k)$. Conditioning on detection replaces that with the normalised density

$$f_k(H_k) = \frac{\lambda(H_k)\, S(m_k)}{\displaystyle\int \lambda(H)\, S(m_k + H - H_k)\, \mathrm{d}H}. \tag{7}$$

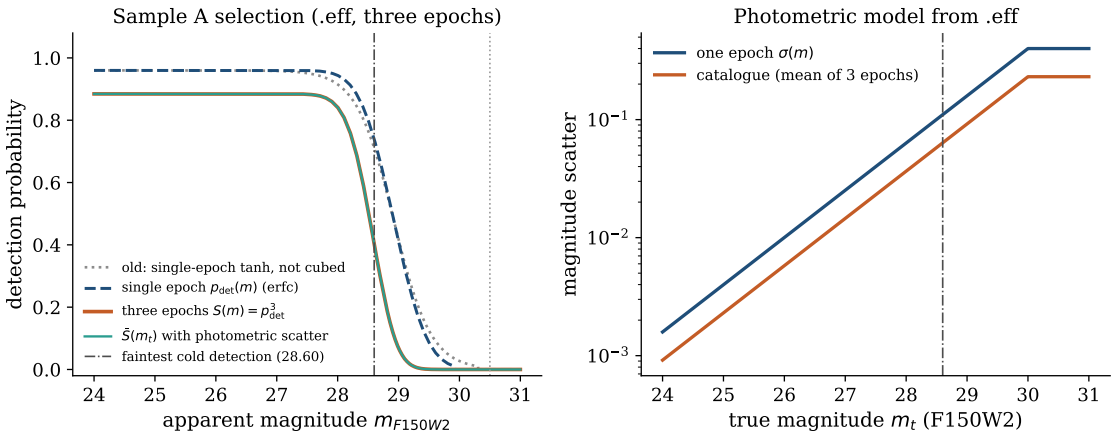


Figure 1: Left: JWST Sample A selection in F150W2, from the three epoch `.eff` files. The single-epoch efficiency $p_{\rm det}$ (dashed) is cubed because Sample A objects were found at all three epochs (solid); $\bar{S}(m_t)$ (Section 4) also averages tracking and characterisation over the photometric noise. The dotted grey curve is the old characterization: the same parameters as a single-epoch tanh, not cubed. The dash-dotted line is the faintest cold detection. Right: the `mag_error` law of the same files, per epoch and for the catalogue magnitude (the mean of three epochs).

The factor $S(m_k)$ in the numerator does not depend on the LF parameters, so it drops out of a maximum-likelihood fit; what matters is that the *denominator* reweights $\lambda(H)$ by the selection function shifted onto the H axis for that object's geometry. If S claims that objects fainter than any detection were readily detectable, the denominator expects them, and the only way the fit can explain their absence is to make $\lambda(H)$ flatter at the faint end. Getting S right is therefore a prerequisite for measuring α (Section 7).

What this buys us. Equation (7) is exactly what you want if the catalogue magnitudes really are truth. It is also the regression check for the measurement-error generalisation: with the photometric model turned off (`measurement="none"` in `hfit.ConditionalLikelihood`, or a `.eff` file with zero `mag_error`), the code reduces to (7) to machine precision.

What this ignores. Near the efficiency drop a few tenths of a magnitude of photometric error is not a small perturbation. Because $\lambda(H)$ rises toward faint H, more objects scatter from faint true magnitudes to brighter measured ones than the other way, so measured magnitudes near the limit are biased bright, and some of the faintest detections are objects that would not have been found at their true magnitude. That is the Eddington bias. Treating catalogue H as exact folds it into the fitted faint end; Section 4 puts it in the likelihood instead of hoping the binning washes it out.

4 Photometric measurement error

4.1 What the survey simulator actually does

OSSSSim does not draw a Gaussian magnitude error and stop. For each plant it draws a measured magnitude from the `magran` model (`numutils.f95` / `ossssim.magerr`):

$$\hat{m} = m_t + \mathrm{offset}(m_t) + \sigma(m_t)\, T_n, \tag{8}$$

where m_t is the true apparent magnitude, $\sigma(m)$ is a piecewise function of six `mag_error=` parameters, offset(m) is a deterministic faint-end shift (objects fainter than a threshold are

measured systematically brighter when the last parameter is negative), and T_n is the mean of $n \in \{1, 2, 3\}$ unit-variance triangular random variables. The integers n are drawn with probabilities `phot_frac` (one, two, or three independent measures averaged). For Sample A the catalogue magnitude is the mean over the three epochs, so `phot_frac= 0 0 1` and the scatter is $\sigma/\sqrt{3}$.

Two ordering facts matter for the likelihood, because they are how the simulator decides detection:

- **Detection efficiency η is applied to the true magnitude m_t.**
- **Tracking and characterisation are applied to the measured magnitude $\hat{m}$.**

If we want the fit to speak the same language as the bias simulation, the likelihood has to respect that split.

4.2 Latent true magnitude

With scatter, the catalogue value $\hat{m}_k$ is no longer a deterministic map from H. True magnitude $m_t = H + \mu_k$ is latent. The contribution of detection k becomes an integral over that latent magnitude (or equivalently over H):

$$f_k = \frac{\displaystyle\int \lambda(H)\, \eta(m_t)\, p(\hat{m}_k \mid m_t)\, \mathrm{d}H \;\times\; \mathrm{track}(\hat{m}_k)\, \mathrm{char}(\hat{m}_k)}{\displaystyle\int \lambda(H)\, \bar{S}(m_t)\, \mathrm{d}H}, \qquad m_t = H + \mu_k. \tag{9}$$

Here $p(\hat{m}_k \mid m_t)$ is the `magran` density (the triangular mixture above), η and track $\cdot$ char are products over epochs, and the denominator uses the *expected* selection at true magnitude m_t,

$$\bar{S}(m_t) = \prod_e \eta_e(m_t)\, \mathbb{E}[\mathrm{track}_e(\hat{m})\, \mathrm{char}_e(\hat{m}) \mid m_t]\,. \tag{10}$$

The expectation is over the same photometric noise that the simulator draws; each epoch draws its own noise, so $\bar{S}$ is a product over epochs.

The factors $\mathrm{track}(\hat{m}_k)\, \mathrm{char}(\hat{m}_k)$ sit outside the H integral in the numerator: they are constants for a fixed catalogue magnitude, so they cancel between detections of equal characterisation and do not move the MLE. Numerically we tabulate

$$\mathrm{NUM}(H) = \eta(m_t)\, p(\hat{m}_k \mid m_t), \qquad \mathrm{DEN}(H) = \bar{S}(m_t) \tag{11}$$

on a fine H grid and replace the integrals by trapezoidal matrix products against $\lambda(H)$.

4.3 Why this form, and not a simpler one

A few alternatives and why we did not stop at them:

Gaussian convolution of the LF. Blurring $\lambda(H)$ with a fixed σ_H is the right move when the noise is on H and selection is flat. Here the noise is on m, selection is a sharp function of m, and η and char see different magnitudes. A single convolution does not capture that.

Forward-model each detection with a Monte Carlo. Drawing many `magran` realisations per likelihood call works, and we use that as a unit test, but it is too noisy and too slow inside MCMC. The grid + analytic mixture kernel is the same integral with stable matrix algebra.

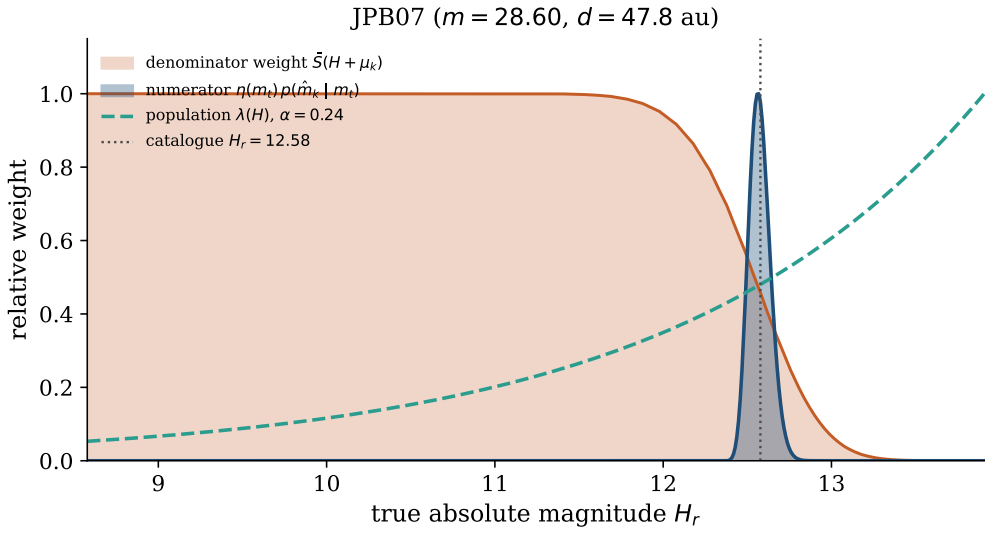


Figure 2: One cold Sample A detection, JPB07. The blue curve is the numerator kernel: which true H values are consistent with the measured magnitude, after folding through η and the photometric model. The orange band is the denominator weight $\bar{S}(H + \mu_k)$, which falls off at the three-epoch limit. The dotted line is the no-error assignment $H = \hat{m} - \mu$. The green dashed curve is the single-exponential population fitted in Section 7. The conditional likelihood is the λ-weighted average of blue, divided by the λ-weighted average of orange.

Put a prior on H_k and sample every object. Fully hierarchical Bayes is attractive and is not ruled out. For the shape-only product likelihood we need now, the per-object latent H integrates out in closed form on the grid, which keeps the fit comparable to the no-error code path and cheap to bootstrap.

The guiding constraint is consistency with OSSSSim: whatever we claim about bias in the HT weights, the shape fit should assume the same measurement process.

5 Distance uncertainty

Absolute magnitude depends on distance. If d is wrong, H is wrong. For short-arc detections the fractional distance error can be several percent, which is several tenths of a magnitude in the distance modulus when $r \sim \Delta \sim d$.

5.1 What we assume

When a per-object distance error σ_d is available (column `d_bary_err` in the detection table), we treat the orbit-fit distance as a measurement of $\ln d$ with

$$\ln d \;\sim\; \mathcal{N}\big(\ln d_{\mathrm{bary}},\; [\ln(1 + \sigma_d/d_{\mathrm{bary}})]^2\big)\,. \tag{12}$$

That is: lognormal errors, fractional width σ_d/d. We are *not* yet folding in a population prior $p(d \mid i)$ from the OSSOS model; that is listed as future work. The present treatment is "the orbit fit said $d \pm \sigma_d$, and we had a flat prior in $\ln d$ before that measurement."

5.2 How the integral is done (without the jargon)

The likelihood (9) now has one more latent variable: distance. We need to average f_k over the distance PDF. Rather than drawing thousands of random distances every time we evaluate the

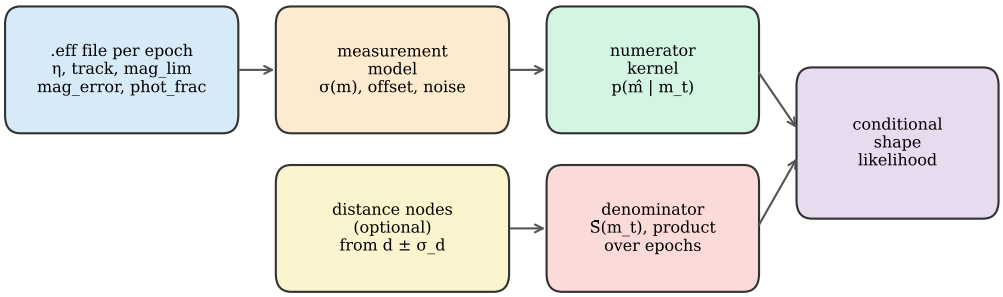


Figure 3: Information flow in the measurement-aware shape fit. The `.eff` file of each epoch supplies both the selection function and the photometric model. Distance nodes (Section 5) are optional and enter as a discrete mixture over geometries.

likelihood, we replace the integral by a short weighted sum

$$f_k = \sum_{j=1}^{J} w_{kj} \, \frac{\mathrm{NUM}_{kj}}{\mathrm{DEN}_{kj}}, \tag{13}$$

where each index j is a single *representative distance* d_j, the weight w_{kj} is how much of the distance PDF that representative carries, and $\mathrm{NUM}_{kj}/\mathrm{DEN}_{kj}$ is equation (9) evaluated at that fixed distance (so μ_k becomes $\mu_k(d_j)$).

The particular distances and weights we use are the classical Gauss–Hermite quadrature nodes for a Gaussian in $\ln d$. In plain language: for a smooth integrand and a roughly Gaussian uncertainty, seven carefully chosen distances already reproduce a 200-point brute-force integral to a part in 10^3 or better. `hfit` uses seven nodes for every detection with a `d_bary_err`; without one, $J = 1$, $d_1 = d_{\mathrm{bary}}$, $w = 1$, and we are back to a fixed geometry.

5.3 What this is not

- It is not a joint orbit–LF fit. We do not re-fit a and e.
- It is not a hierarchical population distance model. Objects do not share a $p(d)$ that knows about the classical belt vs. scattering.
- It does not replace the HT cell assignment. Bias cells are still labelled by the catalogue H; fixing that inconsistency is future work on the normalisation side.

6 Practical fitting

Grid. Everything is tabulated once on an H grid with step $\mathrm{d}H = 0.01$ mag, from 1 mag brighter than the brightest detection to where $\bar{S}$ vanishes for the faintest geometry. A likelihood call is then two matrix–vector products (times J distance nodes).

Measurement model. `survey_debias_hfit` always applies the photometric model of each block's `.eff` files. If a detection carries its own σ_m in a `mag_err` column, that Gaussian width overrides the `.eff` model for the numerator kernel of that object. The denominator still uses the block's model for $\bar{S}(m_t)$, because that is what the survey used when deciding characterisation. The no-error likelihood (7) is available from Python as `ConditionalLikelihood(..., measurement="none")`; the demo in Section 7 uses it as a difference check.

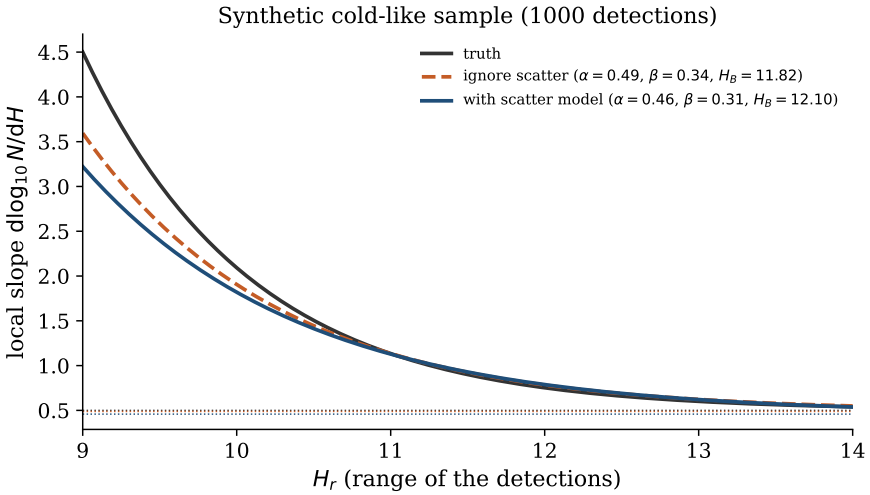


Figure 4: Synthetic check. 1000 detections are drawn from a tapered LF with $(\alpha, \beta, H_B) = (0.5, 0.4, 11.5)$ at cold-belt distances (42–48 au), detected with the Sample A three-epoch $p^3_{\rm det}$ on the true magnitude, and measured with the Sample A photometric model. The detections span $H_r = 10.0$–13.9, so they reach both sides of H_B. Curves are the local slope, equation (3), over that range; dotted lines mark each α. Both fits recover the slope across the detections, and (α, β, H_B) to within about 0.05, 0.1 and 0.6. At Sample A's photometric precision ($\sigma \approx 0.06$ mag at $m = 28.6$ for the three-epoch mean) ignoring scatter changes little. The scatter model matters more for single-epoch catalogues and larger σ.

Inference. We maximise $\sum_k \ln f_k$ (MLE), then run an MCMC with flat priors on $\alpha \in (0.05, 2)$, $\beta \in (0.02, 2)$ and H_B from 5 mag brighter than the grid start to 5 mag fainter than the faintest detection, and a derived H_o from the HT count when a normalisation plot is needed. The fit summary also reports the single exponential $N(< H) = 10^{\alpha(H - H_o)}$ and the $\Delta \ln L$, AIC and BIC between the two. Nonparametric bootstrap of detections *within each block*, with optional log-normal draws of the cell biases, gives a sampling-plus-bias uncertainty that does not pretend $\sum 1/P$ is a known constant.

Separate ridges. With few detections the tapered likelihood can have separate maxima: one with H_B just brightward of the brightest object (the taper as a sharp bright-end cut-off) and one with H_B faintward of the faintest (a slope that flattens across the data, with α at its bound). The MLE in `hfit` starts from a fixed grid of (α, β, H_B) and can stop on the lower one; for the Sample A cold objects it does. Read the MCMC posterior, or a profile likelihood in α as in Section 7, rather than the MLE alone.

Calibration checks. Probability-integral transforms (PITs) of the observed magnitudes under the fitted model should be uniform if the shape and the measurement model are right. Posterior-predictive p-values compare the observed PIT Kolmogorov–Smirnov distance to the distribution expected for uniform deviates of the same sample size (replicates drawn from the model at a posterior θ are uniform by construction, so we do not need to resimulate magnitudes for that particular PPC).

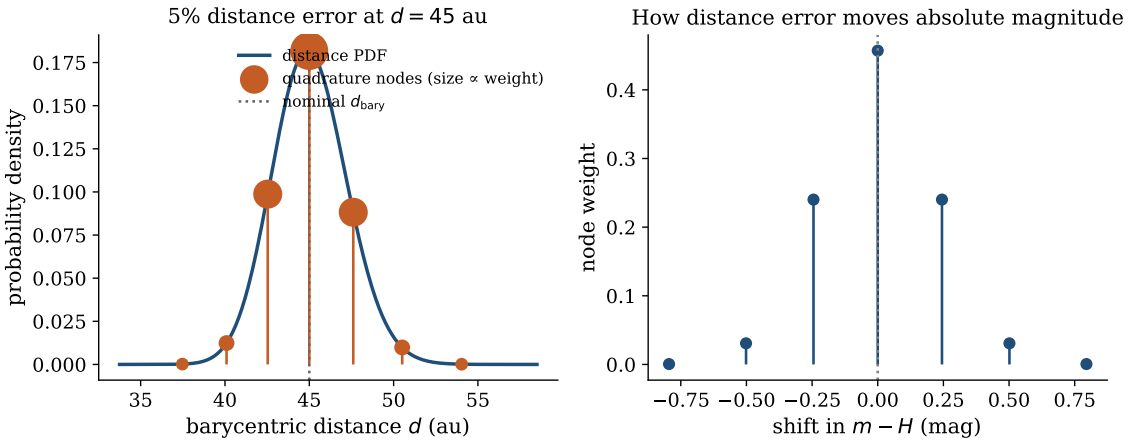


Figure 5: Left: lognormal distance PDF for $d = 45$ au with 5% fractional error, and the seven quadrature nodes (marker size $\propto$ weight). Right: the same nodes expressed as a shift in $m - H$. A 5% distance error is a ~ 0.2 mag uncertainty in absolute magnitude when $r \sim \Delta \sim d$.

7 Results: JWST Sample A, cold objects

7.1 Sample and selection

Eduardo et al. (2026; E26) Sample A is the 20 TNOs found at all three epochs of the GO 1568 NIRCam F150W2 search. We fit only the twelve dynamically cold objects ($38 < d < 55$ au, $i \leq 5°$, `comp = cold`), which span $H_r = 8.59$–13.07 and $m_{F150W2} = 24.47$–28.60. The detections and the selection come from the SurveyCharacterizations survey `characterization/jwst_sample_a/`: `jwst_sample_a.csv` and `epoch{1,2,3}/jwst_sample_a.eff`. Each `.eff` tabulates the single-epoch efficiency of E26 Eq. (8), $p_{\rm det} = (0.96/2)\,\mathrm{erfc}((m-28.92)/0.61)$, and `survey.conf` requires all three epochs, so $S(m) = p_{\rm det}^3$: 0.5 at $m = 28.51$ and 0.4 at 28.60, E26's quoted Sample A limit. The photometric model is $\sigma = 0.11 \times 10^{0.4(m-28.6)}$ per epoch, averaged over three epochs. Figure 6 shows the detections in the (H, m) plane.

7.2 Shape-only fits

Table 1 and Figures 7 and 8 report fits with `hfit.ConditionalLikelihood` under three treatments, plus the old selection for comparison:

1. no measurement error (equation (7));
2. photometric model from the `.eff` files, fixed distances;
3. photometric model plus an illustrative 5% fractional distance error on every object (seven distance nodes);
4. no measurement error with the *old* characterization: the same Eq. (8) parameters as a single-epoch tanh, not cubed.

7.3 Reading the result

Why the earlier version of this note found α at its lower bound. That version applied the *single-epoch* efficiency to a sample that must be found at all three epochs. Uncubed, the selection says an object at $m = 29.0$ is detected about 40% of the time; cubed, 7%. The single-epoch curve predicts about five of twenty Sample A detections fainter than the faintest

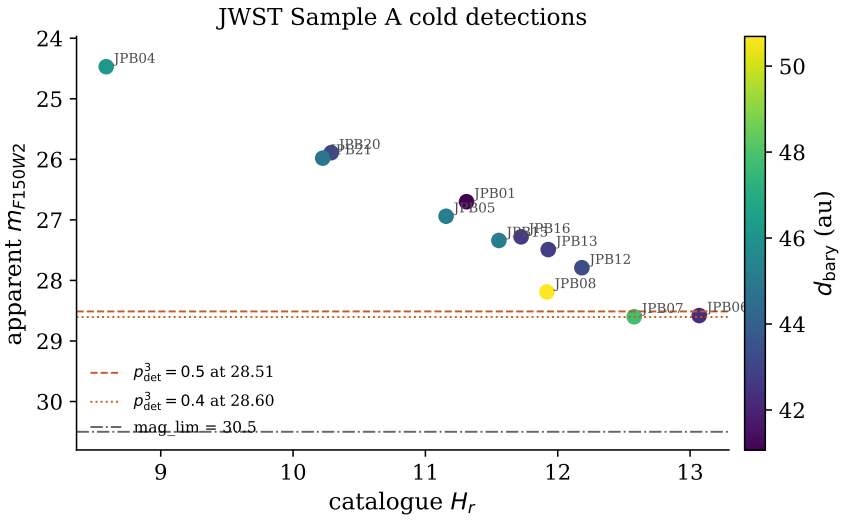


Figure 6: Cold Sample A objects: catalogue H_r vs. apparent m_{F150W2}, coloured by barycentric distance. The lines are where the three-epoch selection $p^3_{\rm det}$ falls to 0.5 and 0.4, and the `mag_lim` of the `.eff` files.

Table 1: Shape-only fits to the twelve cold Sample A objects. Left: the tapered maximum likelihood, refined from the profile in α (a dagger marks a parameter at its prior bound, $\alpha \geq 0.05$, $\beta \leq 2$). Right: the single exponential, with its 68% interval ($\Delta \ln L = -0.5$).

	tapered				single exponential	
treatment	α	β	H_B	$\ln \mathcal{L}$	α	$\ln \mathcal{L}$
no measurement error	0.266	$2.00^\dagger$	8.56	−17.28	$0.24^{+0.10}_{-0.09}$	−17.91
photometric model	0.267	$2.00^\dagger$	8.56	−17.31	$0.24^{+0.10}_{-0.09}$	−17.93
photometry + 5% distance	0.268	$2.00^\dagger$	8.47	−17.55	$0.24^{+0.10}_{-0.09}$	−17.93
old single-epoch S	$0.05^\dagger$	0.21	11.35	−18.49	$0.16^{+0.08}_{-0.08}$	−19.35

one observed (28.60), against 1.6 for $p^3_{\rm det}$ (see the characterization README). Through the denominator of equation (7), the fit can only explain that missing faint tail by flattening $\lambda(H)$ faintward. Since the tapered form's local slope is never below α (equation (3)), it does so by sending α to its bound and placing H_B inside the data, so that the taper provides the rise across the observed range (Figure 8). The taper was working at the bright end, as it always does; it was not eating faint objects. The single exponential, which has no such freedom, gives a lower slope too (0.16).

With the three-epoch selection. The faint-end slope is measured: $\alpha = 0.27$ for the best tapered fit and $0.24^{+0.10}_{-0.09}$ for the single exponential, in line with E26's 0.27 and with the slow rise of the detected histogram. The upper side is well constrained: the profile drops by 0.5 at $\alpha \approx 0.35$ and by 5 at 0.6. The tapered fit gains only $\Delta \ln L = 0.63$ over the single exponential for two more parameters (ΔAIC = 2.7 in favour of the exponential), and it does so by putting H_B at the brightest object (JPB04, $H_r = 8.59$, 1.6 mag brighter than the next) with β at its bound: a cut-off fitted to one object. The taper is not required by these data.

The low side of α is open. A second ridge, α at its bound with $H_B \approx 13.8$ (fainter than every detection), lies only 0.42 below the maximum (0.18 with distance error). On that ridge α describes $H > 13.8$, where there are no detections; across the data the local slope falls from 0.6

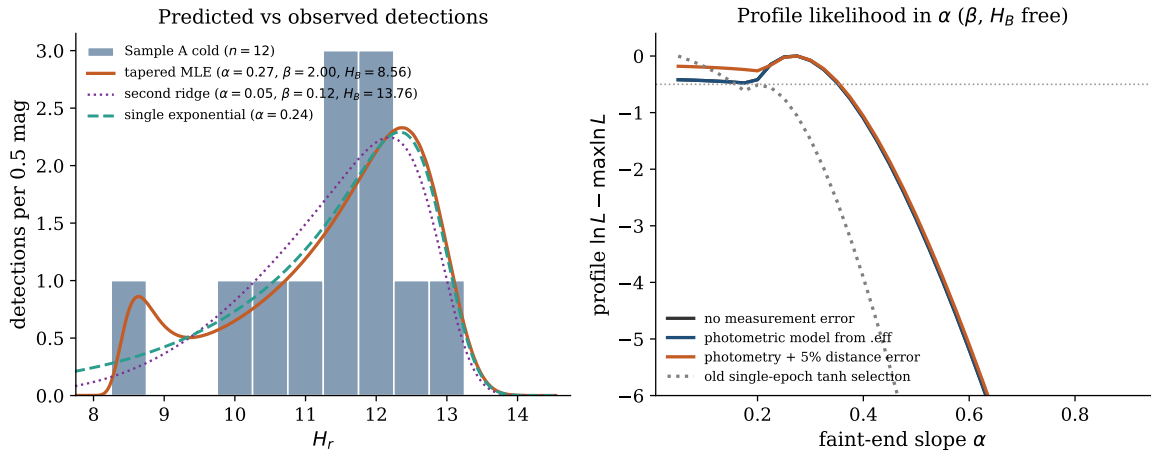


Figure 7: Left: catalogue H_r of the cold objects and the predicted distribution of detected H (photometric model) for the best tapered fit, the second ridge, and the single exponential. Right: profile likelihood in α, maximising over (β, H_B) at each α. With the three-epoch selection all treatments peak at $\alpha \approx 0.27$ and fall by 5 at $\alpha = 0.6$; the low side is open (the second ridge reaches the bound at $\Delta \ln L = -0.42$). With the old single-epoch selection the profile rises to the bound.

to 0.2. An MLE at $\alpha = 0.05$ is then a statement that the faint end is beyond the data, not that the population is flat. `hfit`'s grid-started MLE lands on this ridge for these data (Section 6).

Measurement error. At Sample A's precision ($\sigma \approx 0.06$ mag at $m = 28.6$ for the three-epoch mean) the photometric model moves $\ln L$ by 0.03 and the parameters by less than 0.01. A 5% distance error flattens the low-α side of the profile slightly and moves the best fit by less than 0.1 mag in H_B.

The honest summary for the Sample A cold objects is therefore:

- With the selection the survey actually applied (three epochs), the data give a faint-end slope of about 0.25: well bounded above ($\Delta \ln L = -0.5$ at about 0.35, -5 at 0.6) and soft below.
- Twelve objects do not constrain a bright-end taper.
- The *machinery* (no-error vs. photometric vs. distance mixture) is in place and recovers a known tapered LF from a synthetic sample that spans H_B (Figure 4).
- A useful next measurement is the HT-normalised cumulative counts with bootstrap bias errors, which ask a different question (yield, not shape), and/or a joint fit with a larger ground-based sample sharing the same $\lambda(H)$, which would reach brightward of any taper.

Numbers in Table 1 were written by `make_figures.py` to `results/jwst_shape_mle.csv` and `results/jwst_shape_notes.txt`. Re-run that script after changing the `.eff` files or the detection table.

8 What we are still not doing

- Colour and per-block zero-point nuisance parameters with priors. Those are shared offsets, degenerate with H_o and H_B.

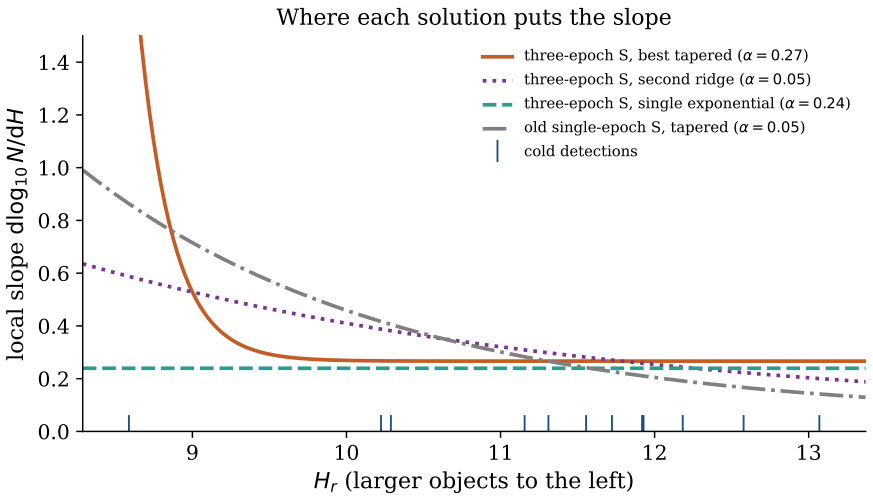


Figure 8: Local slope, equation (3), of each solution across the cold detections (ticks). Larger objects are to the left. The best tapered fit is a constant $\alpha \approx 0.27$ with the taper acting only brightward of the brightest object. The second ridge and the old-selection fit put α at its bound and use the taper to make the slope fall faintward across the data.

- A population distance prior $p(d \mid i)$ from the OSSOS model, needed before distance marginalisation is really Bayesian for `model_ae` surveys.
- An extended Poisson likelihood for the normalisation, replacing the HT count. Until then, HT cells are still labelled by measured H while the bias plant uses true H.

9 Reproducing the figures

The script needs the SurveyCharacterizations repository checked out beside this one. From the repository root, with `numpy`, `scipy`, and `matplotlib` installed:

```
python docs/lf_shape_fit/make_figures.py
pdflatex -output-directory=docs/lf_shape_fit \
         docs/lf_shape_fit/lf_shape_fit.tex
```

The script imports `survey_debias` from this checkout's `src/`. Figure PDFs land in `docs/lf_shape_fit/figures/`; the fit table and notes land in `docs/lf_shape_fit/results/`.